

Contactless Blood Pressure Monitoring with mmWave Radar

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Abstract—The monitoring of blood pressure is critical for the prevention, diagnosis and treatment of cardiovascular diseases. However, existing methods require physical contact between human body and sensor, which are not suitable for long-term monitoring. In this paper, we propose a contactless blood pressure monitoring system, mmBP, using millimeter wave radar. Specifically, we first separate signals reflected from different spatial locations by coherently combining the signals on different antennas. Then, we locate and extract the arterial pulse using convolutional neural network (CNN) assisted template matching with location tracking. Finally, we design an encoder-decoder neural network to derive the blood pressure information from the extracted signal. Experimental results on 20 subjects show that the measurement deviation rate is 9.00% and 3.69% for systolic and diastolic blood pressure, which demonstrates the feasibility and effectiveness of the proposed system.

Index Terms—Millimeter wave radar, blood pressure, deep learning, contactless monitoring

I. INTRODUCTION

Blood pressure is one of the most important indicators for the diagnosis and treatment of cardiovascular diseases. Measuring the blood pressure continuously in operation could remind doctor of the patient's physiological state and ensure the life safety of patient, while monitoring the blood pressure in daily life could effectively avoid emergencies and provide better medical care for patients [5]. However, existing methods for blood pressure measurement require cuffs [17]–[20] or other contact-based sensors [6], [21]. The sensors have to be in contact tightly with human skin, which is uncomfortable for patients and unpractical for long-term monitoring.

To measure blood pressure contactlessly, camera-based solutions have been proposed, where photoplethysmogram (PPG) signals [6], an optical signal proportional to the blood-volume change, have been utilized to extract arterial pulse. Then by analyzing the waveform of arterial pulse, blood pressure can be estimated. However, the camera-based methods highly depend on the lighting conditions. When the lighting conditions are not good or there is interference from other lighting sources, the performance would degrade rapidly. Besides, the use of camera would introduce severe privacy concerns, which further limits the practical deployment.

To overcome the limitations of optical cameras, millimeter wave (mmWave) radars have been utilized to extract blood pressure information contactlessly. Since the arterial pulse wave would introduce vibration on body surface, it is feasible to extract arterial pulse wave using radar [2]. Compared with camera, mmWave radars do not depend on lighting conditions, can penetrate obstacles, and are privacy friendly. More importantly, mmWave radars are sensitive to small movements, such as human breath, heartbeat and arterial pulse.

In this paper, we propose a contactless blood pressure monitoring system using mmWave radar, denoted as mmBP. The core idea of mmBP is to extract cardiovascular information from the signal variation caused by the arterial pulse on the surface of human body. Specifically, we first separate signals reflected from different spatial locations by coherently combining the signals on different antennas [7]. Then, we locate the arterial pulse signal using CNN assisted template matching and dynamically track the arterial pulse in a small range. Compared with existing methods which simply pick peaks in the spectrum, the CNN based template matching could better capture fine-grained signal variation. Finally, we design an encoder-decoder neural network to derive the cardiovascular information from the extracted signal to estimate blood pressure. The design of the neural network is inspired by the success of attention mechanism, which could focus on the meaningful information in the extracted signal automatically. To evaluate the proposed mmBP, we collect data on 20 objects and the measurement deviation rate is 9.00% and 3.69% for systolic and diastolic blood pressure, which demonstrates the feasibility and effectiveness of the proposed system.

II. METHOD

A. Radar Signal Acquisition

In this paper, we use the frequency modulated continuous wave (FMCW) mmWave radar, which utilizes frequency ramping of a transmit signal to detect the position of a target. The phase of the beat signal, i.e., the intermediate frequency (IF) signal, at the target location is utilized to extract the motion of the target. This phase-based approach can measure tiny motion and has been utilized successfully in other vital sensing applications [2] [4]. Specifically, we use the Texas

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Instruments (TI) AWR1843 [24], which consists of 3 transmit antennas and 4 receive antennas operating in 77-81 GHz band.

B. Spatial Separation

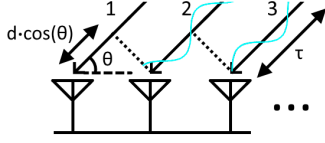


Fig. 1: The signal arrives at the antenna array with the AOA of θ and the TOF of τ . And d denotes the distance of every antenna.

To separate the signals from different spatial locations, we coherently combine the signals from different antennas. Considering the signal transmitted or reflected from angle of arrival (AOA) θ , time of flight (TOF) τ , the relative phase shift of this signal on adjacent antennas can be expressed as [3]

$$\Phi(\theta) = e^{-\frac{j2\pi d \cos \theta}{\lambda}}, \quad (1)$$

while the phase shift on adjacent samples is given by

$$\Phi(\tau) = e^{-j2\pi k_s \tau}, \quad (2)$$

where k_s denotes the slope of the frequency modulation, and λ denotes the electromagnetic wave wavelength.

By compensating the phase shift and adding the signals on different antennas and samples, the signals from AOA θ and TOF τ would superimpose coherently while the signals from other locations would be suppressed. In this way, we separate the signals into grids of angles and distances. As the phase shift of the echo corresponds to the displacement of the object, we extract the phase of signal to characterize the arterial pulse motion on the skin surface [2].

Through the aforementioned procedures, most of the environment reflections have been suppressed. However, small movements and other physiological signals of the subject can still affect the quality of the arterial pulse. To alleviate this problem, we have noted that the frequency of arterial pulse would be higher than 0.5Hz. Hence, we further utilize an FIR filter to remove the irrelevant signals.

C. Locating and Dynamic Tracking

The next step of the proposed system is to locate and dynamically track the spatial grid corresponding to arterial pulse. To achieve this, we adopt a CNN based template matching method [8]. Compared with simply picking the signal frequency corresponding to arterial pulse using FFT, the proposed method could better capture the variation of the arterial pulse signal in time domain [8].

The signal is divided into pieces D_i containing at least one arterial pulse and we adopt the template T for matching. The D_i are all equal length and the length is long enough. The objective function of the template matching is to minimize the error E_i which is given by

$$E(i) = |D_i - T|^2 = |D_i|^2 + |T|^2 - 2|D * T|. \quad (3)$$

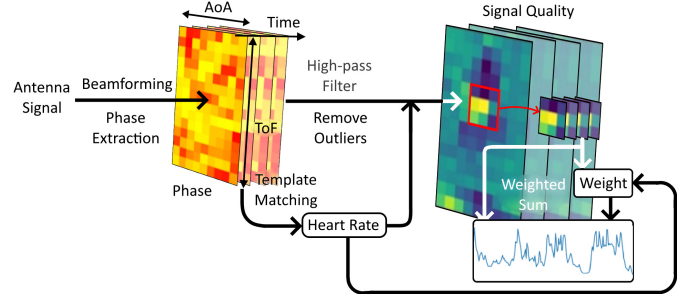


Fig. 2: Extract and evaluate the signal to calculate arterial pulse.

In the proposed framework, the template can be regarded as an 1D convolution kernel, and the template matching can be regarded as performing convolution. As a result, $|T|^2$ can be seen as the l_2 regularization of the convolution kernel, and $|D * T|$ can be seen as the convolution output. Hence, we define the loss function as follows:

$$Loss_0 = L_2(k) + \frac{\mu}{\sum_i^N \text{conv}(D_i, k)}, \quad (4)$$

where L_2 is L_2 regularization function, k is the convolution kernel, μ is super parameter, $\text{conv}()$ is the convolution operation, N is the number of time windows. Such an optimization problem can be solved with gradient descent (GD).

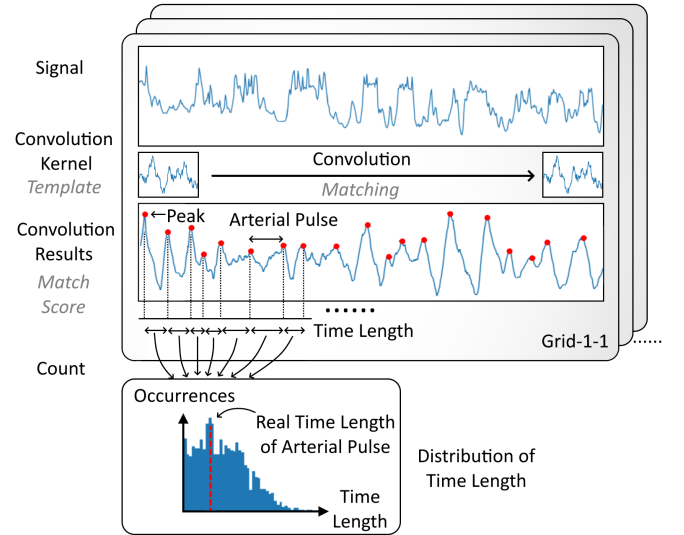


Fig. 3: Extract heart rate using template matching.

We then utilize the convolution kernel k to convolve with the whole signal, in other words, to slide on the signal and perform template matching. When the template is perfectly matched, there would be peaks in the output. Between the two adjacent peaks is a beat of arterial pulse, and the time between two peaks is the time of a beat of arterial pulse. We count the occurrences of the time length of these beats, and select the most frequent time length as the real time length of arterial pulse.

Note that the subjects may not remain stationary all the time, which would lead to the location variation of the arterial pulse. To prevent such a problem, we extract the signal from a 3×3 grid around the location of the arterial pulse and calculate the weighted sum of signals from these grids. The weight is calculated by evaluating the signal of 3×3 grid with the extracted heart rate. The weight is defined as follows:

$$w = \text{Softmax}(1/|f_p - f_i|), i = 1, 2, \dots, 9 \quad (5)$$

where f_i is the frequency of the signal of i 'th grid, and f_p is the frequency of arterial pulse. For $\{x_1, x_2, \dots, x_N\}$, $\text{Softmax}(x_i) = e^{x_i} / \sum_{j=1}^N e^{x_j}$

As the weight is calculated dynamically, even if there is disturbance on position of arterial pulse, the grid of arterial pulse can still achieve the maximum weight. Moreover, due to the energy leakage of beamforming, there may also be arterial pulse signals around the real position of arterial pulse. By adding these arterial pulse signals, the arterial pulse signal is enhanced and the noise is further suppressed, i.e., the SNR is improved.

D. Cardiovascular Feature Extraction

It has been shown that there exists correlation between the arterial pulse and blood pressure [15], however the corresponding relationship is complicated and difficult to be explicitly characterized. Thus, in this paper, we propose to use deep neural network to learn such a relationship.

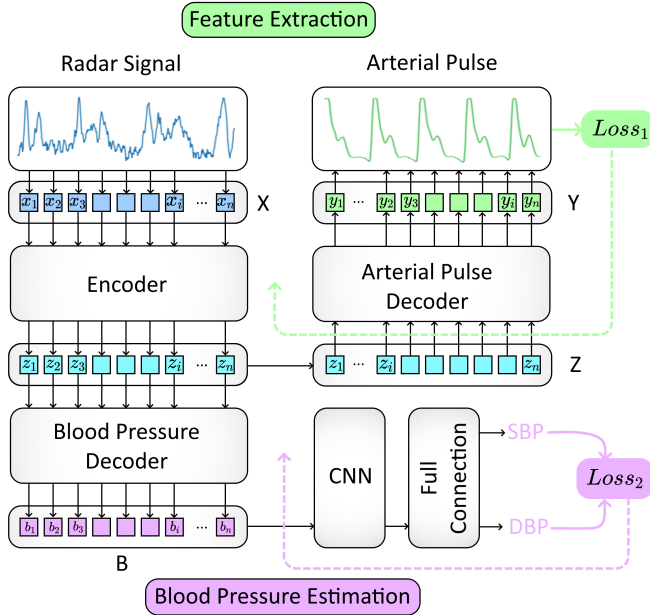


Fig. 4: Extract information with encoder-decoder network and predict blood pressure with CNN.

1) *Feature Extraction*: As shown in Fig. 4, we adopt an encoder-decoder architecture to extract cardiovascular feature. The encoder takes the radar signals as input and output

the cardiovascular features. To extract cardiovascular features robustly, we introduce the arterial pulse as supervision. Specifically, we design an arterial pulse decoder to rebuild the arterial pulse from cardiovascular features. Assuming the range of the amplitude of arterial pulse wave is from 0 to R , and the length of arterial pulse is S . The loss function is designed as the sum of cross entropy of the rebuilt waveform and the reference waveform at every moment, which is defined as

$$\text{Loss}_1 = - \sum_i^S \sum_j^R \text{ref}_j^i \log(y_j^i), \quad (6)$$

where y^i is the possibility distribution of amplitude, y_j^i is the possibility of the amplitude at moment i equals j , and the ref^i is defined as a one-hot vector at moment i .

As shown in Fig. 7, there are some critical parts in the arterial pulse waveform such as trough, crest, and dicrotic notch, which deserves more attention. Thus, in this paper, we utilize a modified transformer [1] with self-attention layers to automatically focus on the importance parts of the signal. Moreover, such a network architecture can further aggregate the information from different positions, which could be utilized to combine some parts to infer more information, such as $H_1, H_2, H_3, W_1, W_2, W_3$ in Fig. 7.

E. Blood Pressure Estimation

We design our network as a dual-task architecture including pulse waveform reconstruction and blood pressure estimation, where the task of reconstructing the pulse waveform is used to assist the cardiovascular feature extraction for blood pressure estimation task. We use a transformer-based decoder to decode the extracted cardiovascular features. To further extract high-level semantic information, we add several CNN layers after the blood pressure decoder, as showed in Fig. 4, where the residual mechanism is adopted to avoid the vanishing gradient problem [11]. Finally, we use a full collection layer to estimate blood pressure, where the outputs are the systolic and diastolic blood pressure. The loss function is defined as mean squared error (MSE) as below

$$\text{Loss}_2 = (BP_S - BP'_S)^2 + (BP_D - BP'_D)^2, \quad (7)$$

where the BP'_S and the BP'_D are ground-truth systolic and diastolic blood pressure, while the BP_S and the BP_D are the estimations of systolic and diastolic blood pressure.

III. EXPERIMENTAL RESULTS

We collect data from 20 volunteers with different genders and ages at different time and environments. Totally, 15890 frames of signals with length of around 5s have been collected.

The radar data is collected with TI AWR1843 [24] evaluation board. The ground-truth blood pressure is estimated with sphygmomanometer, which has been widely applied in medical diagnosis. The experimental setting is shown in the Fig. 5 (left). To improve the quality of data set, we adopt wave absorbing sponge to cover other parts while only wrist

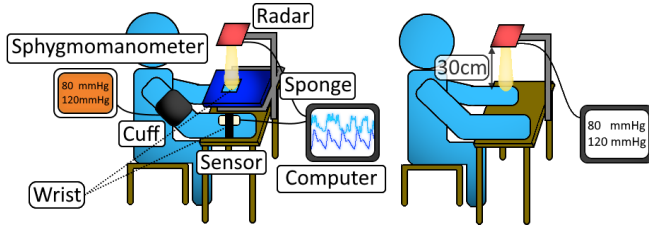


Fig. 5: Experimental setting (left) and setting in use (right).

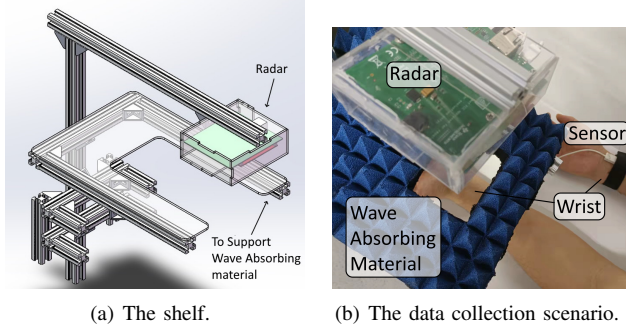


Fig. 6: The shelf to support the radar and electromagnetic wave absorbing material (a) and data collection scenario (b).

can reflect signal to radar to eliminate the interference from other body parts. To avoid the self-interference problem, the radar is placed about 30cm away from the wrist. We designed a shelf in the Fig. 6(a) to support the radar and the sponge.

Since the arterial pulse waveform is used as the supervision in the encoder-decoder network, we record it synchronously with HK-2000H arterial pulse sensor [25], a wearable device on subject's wrist. In order to avoid interfering with radar signal collection, we put the arterial pulse sensor and radar at different sides of wrists. To void the error caused by time misalignment of two channels of signal, we record timestamps at each sampling point to align them. After radar and arterial pulse data are collected, we measure reference blood pressure with a sphygmomanometer. Once the model has been trained, we only use radar to collect the signal and estimate the blood pressure with the trained model, as shown in the Fig. 5 (right).

The sampling rate of arterial pulse sensor and radar is 50Hz. To ensure data quality, we drop the first 30 seconds data, and the subjects are required to rest for a while before data collecting. During data collecting, subjects are required to avoid mood swings and actions. Hence, we can also assume that the blood pressure keeps invariant, and the blood pressure reading of sphygmomanometer is correct. And every time we collect sensor and radar data continuously on one subject for around 3 minutes (not counting the dropped 30 seconds), blood pressure will not fluctuate much.

In the locating phase, we use one minute of data to calculate the position of arterial pulse, and update the position every minute. Then we use the data window of around 5 seconds (256 sampling points) for tracking and blood pressure estimation.

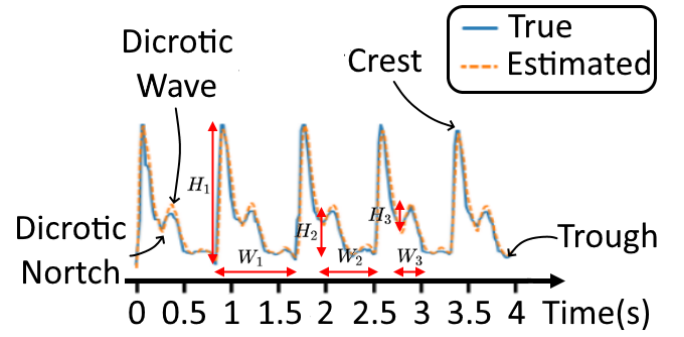


Fig. 7: Estimation of arterial pulse waveform and the real waveform. And important features ($W_1, W_2, W_3, H_1, H_2, H_3$) of arterial pulse.

We update tracking and blood pressure every second, in other words, the window slides 50 sampling points at a time. In the processing of beamforming, the space are cut into 7×14 (width $\times$ height) grids, as shown in Fig. 2. In the template matching, the size of template is 151 (about 3 seconds), and the last time of one arterial pulse is about 1 second, which is enough for the template to cover a complete arterial pulse.

As for the encoder-decoder transformer network, the configuration is set as follows. The encoder and decoders are double-layer. The x_i, y_i, z_i in Fig. 4, the inputs and outputs of encoder and decoders, are 256-dimension vectors, the same as the resolution of ADC (analog to digital conversion) of the arterial pulse sensor. The hidden dimension of the model is 64, with 8 headers. We employ the drop out mechanism, the drop rate is 0.1.

The CNN is stacked up with residual blocks. Our residual block contains 2 convolution layers with ReLU function, and the kernel size is 3, and padding is 1. And we add jump link at every block. Assuming that the input of the block is x , then the out block is

$$x_{out} = x + \text{ReLU}(\text{conv}_2(\text{ReLU}(\text{conv}_1(x)))) \quad (8)$$

where conv_1 and conv_2 are convolution layers.

We use down sampling layer to replace the pooling layers in CNN, and the down sampling layers can be implemented by convolution. The kernel size of the convolution is 2, the stride is 2, and with no padding. The feature size can be halved through the down sampling layer.

The CNN structure consists of alternating two residual blocks and down sampling layers. Every time features pass through a down sample layer, the number of channels for features double. As mentioned above, the input of CNN ($b_1, b_2, \dots, b_n$ in Fig. 4) is a $256 \times 256 \times 1$ tensor. And after 3 down sampling layers and 6 residual blocks, the output is a $32 \times 32 \times 8$ tensor. Then we compress the tensor to a 8192×1 tensor. After the first full connection layer, the feature is mapped to 128×1 , and the second full connection layer maps the feature to 2×1 , which corresponds the systolic and diastolic blood pressure, respectively.

In the process of encoder training, the waveform of the arterial pulse is estimated as an intermediate in our multitask model, as showed in Fig. 7, The features such as dicrotic waves are exactly rebuilt, and the average deviation rate between the estimated waveform and true waveform is 3.6%, which is close to the ground truth. This result indicates that encoder do extract the physiological information to reconstruct the arterial pulse waveform and the information to deduce blood pressure.

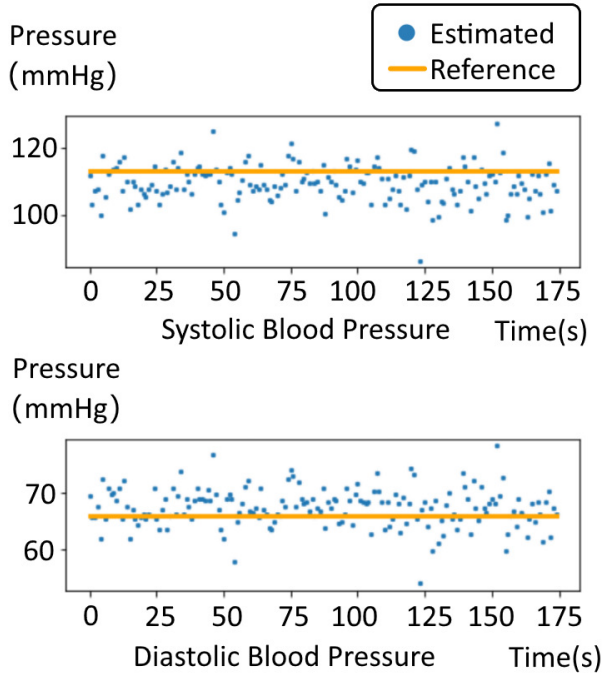


Fig. 8: 3 minutes estimation and the reference of blood pressure of an object.

As our final task, the estimated blood pressure and the reference are showed in Fig. 8. The average deviation of systolic blood pressure is 9.87mmHg, and the average deviation of diastolic blood pressure is 2.49mmHg. The average deviation rate of systolic and diastolic blood pressure is 9.00% and 3.69%, which can satisfy the Association for Advancement of Medical Instrumentation (AAMI) protocol [26].

IV. CONCLUSION

In this paper, we proposed mmBP, a contactless blood pressure monitoring system using millimeter wave radar. The mmBP system separated human reflections from radar signals based on beamforming. Then, it located and extracted pulse signal using CNN based template matching. Finally, an attention based encoder-decoder neural network was designed to derive the blood pressure value. Experimental results on 20 subjects showed that the measurement deviation rate was 9.00% and 3.69% for systolic and diastolic blood pressure, which demonstrated the great application potential of the proposed system.

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